

# NF-UQ-NIDS

## Exploratory Data Analysis Report

|            |         |                |                 |
|------------|---------|----------------|-----------------|
| 11,994,893 | 15      | 0              | 10.55%          |
| Flows      | Columns | Missing values | Duplicates      |
| 76.77%     | 23.23%  | 21             | 4               |
| Benign     | Attack  | Class labels   | Source datasets |

### Purpose

To characterize data quality, class balance, network behavior, feature distributions, correlations, and preprocessing requirements before training a Deep Reinforcement Learning based intrusion detection system.

Prepared from the completed Jupyter Notebook EDA workflow

# 1. Executive Summary

NF-UQ-NIDS contains nearly 12 million NetFlow records from four benchmark intrusion-detection datasets. The analysis found no missing values, but approximately 10.55% exact duplicate rows. Binary labels are moderately imbalanced, while individual attack categories are highly imbalanced. Numerical traffic features are strongly right-skewed and contain large but plausible values. No numerical feature pair is redundant at an absolute correlation above 0.90. IP address columns have high cardinality and are environment-specific, so they should be removed before model training to reduce memorization and improve generalization.

## Primary modeling implication

The dataset is suitable for binary and zero-day IDS experiments, but preprocessing must preserve the Attack field for holdout design, avoid data leakage, scale numeric features, and explicitly handle duplicates and class imbalance.

# 2. Dataset Structure

| Feature                    | Type    | Meaning                          | Decision                  |
|----------------------------|---------|----------------------------------|---------------------------|
| IPV4_SRC_ADDR              | String  | Source IPv4 address              | Drop for modeling         |
| L4_SRC_PORT                | Integer | Transport-layer source port      | Keep                      |
| IPV4_DST_ADDR              | String  | Destination IPv4 address         | Drop for modeling         |
| L4_DST_PORT                | Integer | Transport-layer destination port | Keep                      |
| PROTOCOL                   | Integer | IP protocol number               | Keep                      |
| L7_PROTO                   | Float   | Layer-7 protocol identifier      | Keep                      |
| IN_BYTES                   | Integer | Incoming byte count              | Keep                      |
| OUT_BYTES                  | Integer | Outgoing byte count              | Keep                      |
| IN_PKTS                    | Integer | Incoming packet count            | Keep                      |
| OUT_PKTS                   | Integer | Outgoing packet count            | Keep                      |
| TCP_FLAGS                  | Integer | Aggregated TCP flag bitmask      | Keep                      |
| FLOW_DURATION_MILLISECONDS | Integer | Flow duration in milliseconds    | Keep                      |
| Label                      | Integer | 0 benign / 1 attack              | Target                    |
| Attack                     | String  | Attack category                  | Keep for zero-day split   |
| Dataset                    | String  | Original source dataset          | Analysis / stratification |

### 3. Data Quality Assessment

| Check            | Result                                 | Interpretation                                                             |
|------------------|----------------------------------------|----------------------------------------------------------------------------|
| Dataset shape    | 11,994,893 rows x 15 columns           | Large-scale dataset; full-memory analysis was feasible with available RAM. |
| Missing values   | 0 in every column                      | No imputation is required.                                                 |
| Exact duplicates | 1,265,854 rows (10.5533%)              | High enough to investigate before training.                                |
| Data types       | Strings, integers, one float feature   | Types are consistent with the dataset schema.                              |
| Numeric ranges   | No obviously impossible range detected | Extreme values should not be removed automatically.                        |

**Duplicate caution**

Exact duplicates may represent repeated genuine flow behavior or dataset-construction artifacts. Their class and source distribution should be checked before deciding whether to remove them. Duplicate leakage across train and test must be prevented.

## 4. Class and Source Composition

### Binary Label Distribution

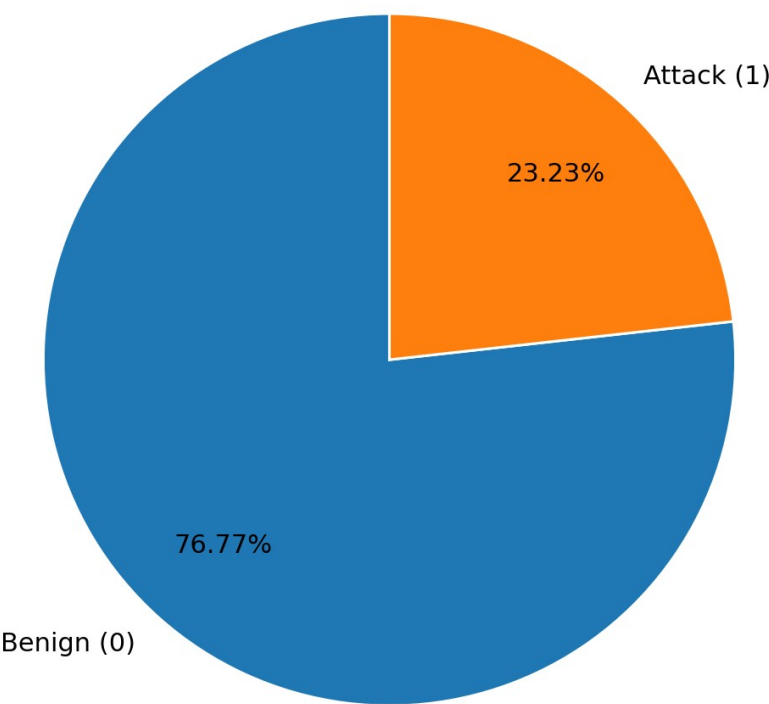


Figure 1. Binary label distribution: benign traffic is the majority class, but attack traffic still forms a substantial 23.23%.

| Label      | Count     | Share  |
|------------|-----------|--------|
| Benign (0) | 9,208,048 | 76.77% |
| Attack (1) | 2,786,845 | 23.23% |

**Baseline warning**

A classifier that predicts every record as benign would achieve 76.77% accuracy while detecting no attacks. Accuracy must therefore be reported with precision, recall, F1-score, ROC-AUC, and a confusion matrix.

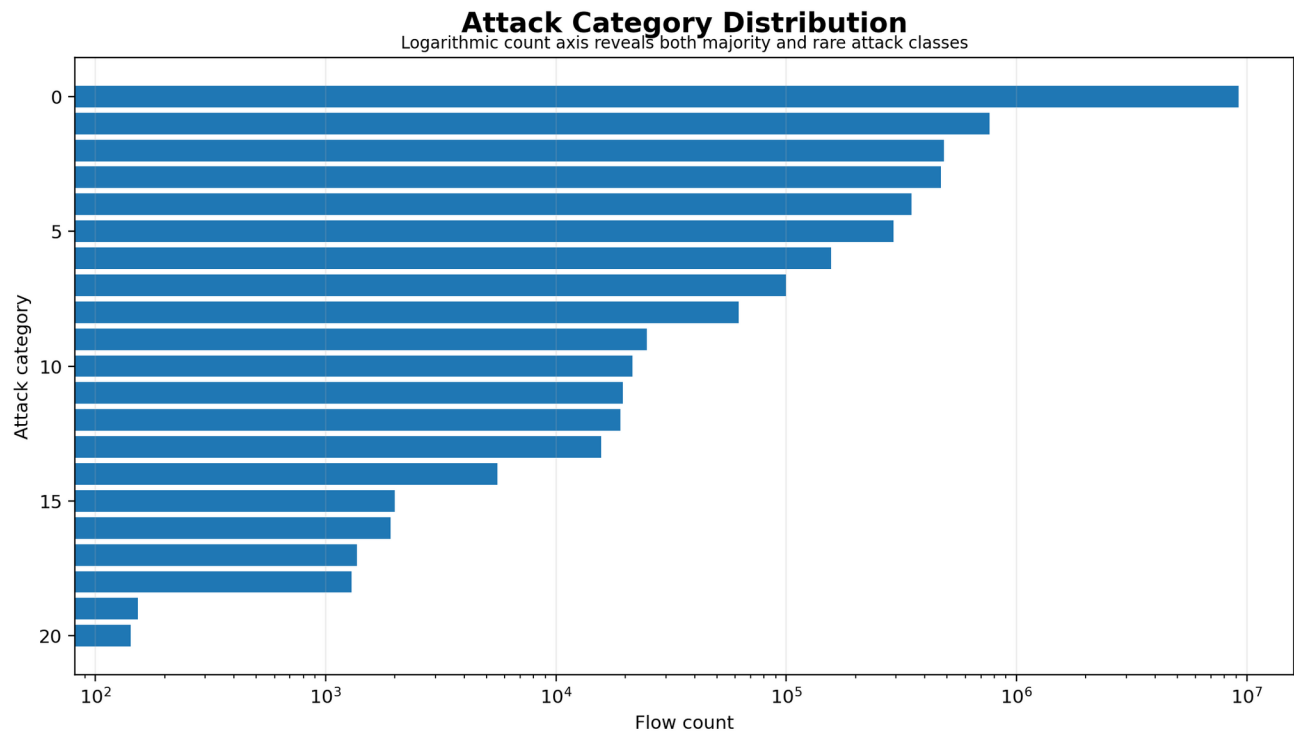


Figure 2. Attack-category distribution. The log axis exposes the severe long-tail imbalance, from 763,285 DDoS flows down to 142 ransomware flows.

| Attack class   | Count     | Dataset share |
|----------------|-----------|---------------|
| Benign         | 9,208,048 | 76.766%       |
| DDoS           | 763,285   | 6.363%        |
| Reconnaissance | 482,946   | 4.026%        |
| injection      | 468,575   | 3.906%        |
| DoS            | 348,962   | 2.909%        |
| Brute Force    | 291,955   | 2.434%        |
| password       | 156,299   | 1.303%        |
| xss            | 99,944    | 0.833%        |
| Infiltration   | 62,072    | 0.517%        |
| Exploits       | 24,736    | 0.206%        |
| scanning       | 21,467    | 0.179%        |
| Fuzzers        | 19,463    | 0.162%        |
| Backdoor       | 19,029    | 0.159%        |
| Bot            | 15,683    | 0.131%        |
| Generic        | 5,570     | 0.046%        |

|            |       |        |
|------------|-------|--------|
| Analysis   | 1,995 | 0.017% |
| Theft      | 1,909 | 0.016% |
| Shellcode  | 1,365 | 0.011% |
| mitm       | 1,295 | 0.011% |
| Worms      | 153   | 0.001% |
| ransomware | 142   | 0.001% |

Zero-day design implication

Very rare classes such as Worms and ransomware provide too few samples for stable standalone evaluation. Mid-sized categories such as Backdoor, Theft, or Shellcode are more practical holdout candidates, depending on the research hypothesis.

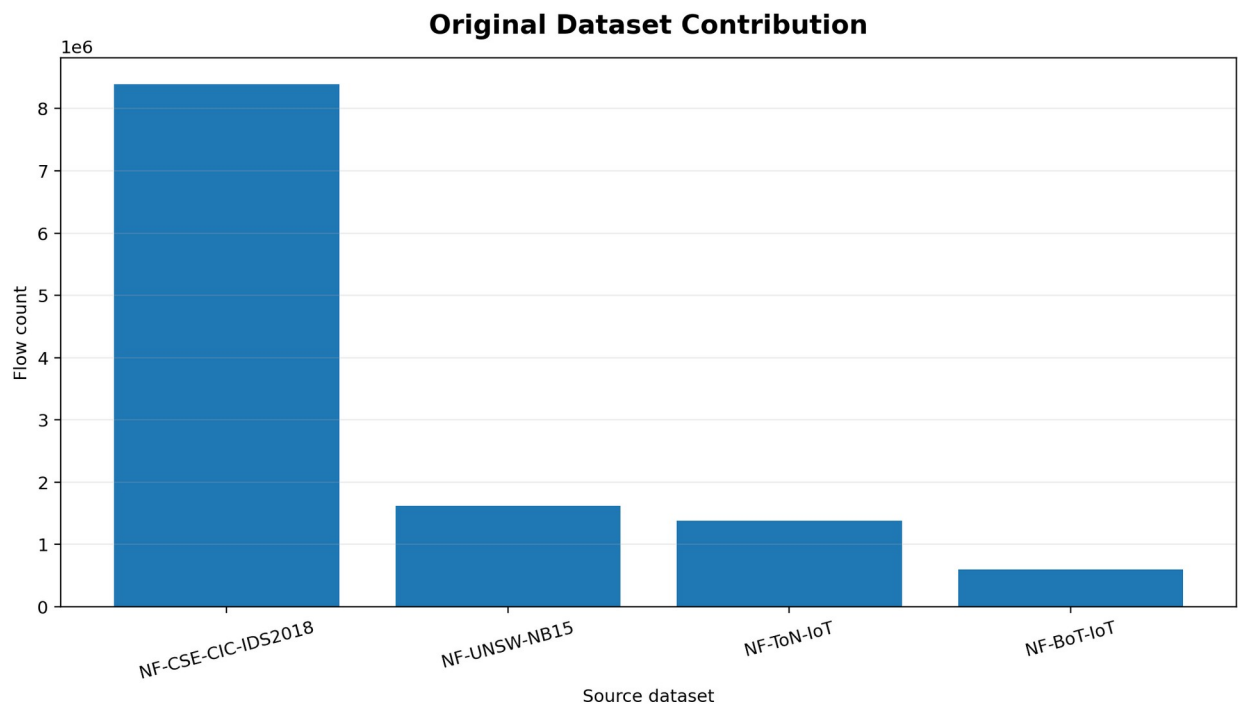


Figure 3. Contribution of the four original datasets. NF-CSE-CIC-IDS2018 contributes almost 70% of all flows.

| Source dataset     | Count     | Share  |
|--------------------|-----------|--------|
| NF-CSE-CIC-IDS2018 | 8,392,401 | 69.97% |
| NF-UNSW-NB15       | 1,623,118 | 13.53% |
| NF-ToN-IoT         | 1,379,274 | 11.50% |
| NF-BoT-IoT         | 600,100   | 5.00%  |

Generalization caution

Aggregate performance may be dominated by NF-CSE-CIC-IDS2018. Reporting results by source dataset can reveal whether the model generalizes across environments rather than merely fitting the largest source.

5. Network Traffic Composition

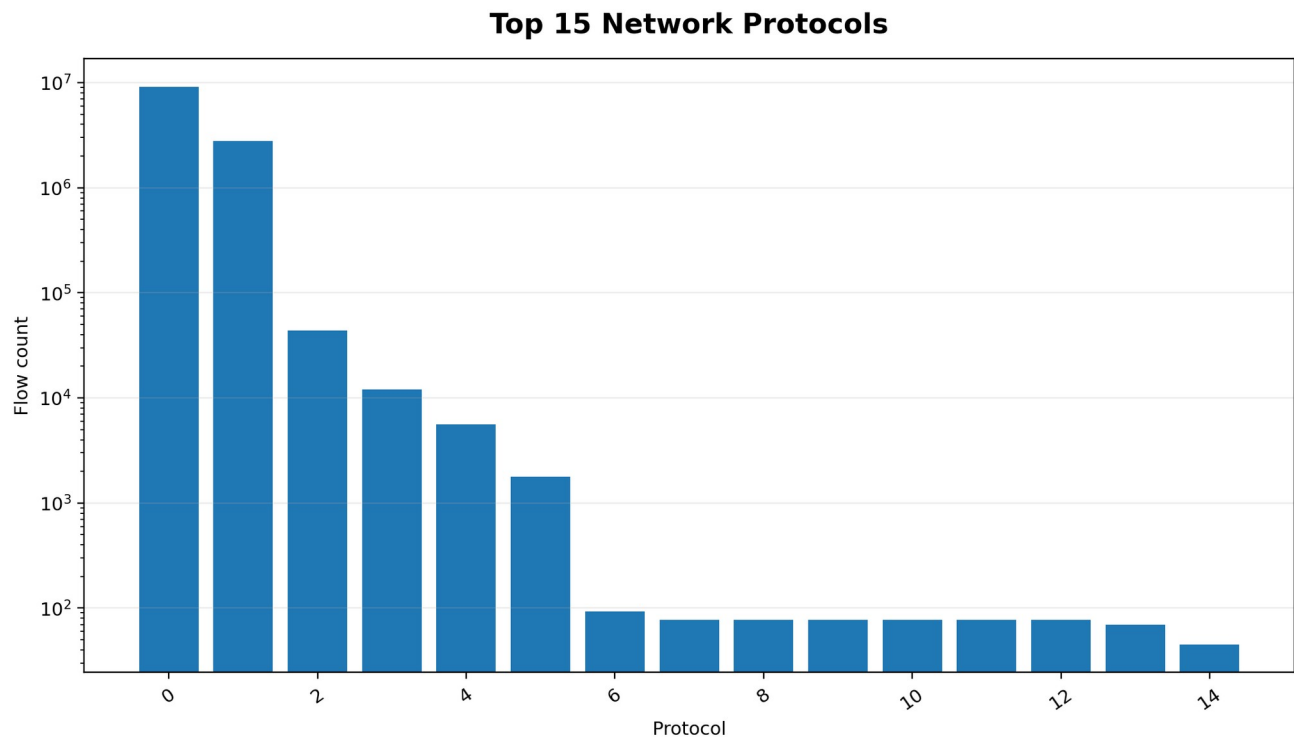


Figure 4. Protocol distribution. TCP and UDP account for more than 99% of all flows; the logarithmic axis makes rare protocols visible.

| Protocol    | Count     | Share   |
|-------------|-----------|---------|
| TCP (6)     | 9,151,668 | 76.296% |
| UDP (17)    | 2,769,768 | 23.091% |
| ICMP (1)    | 43,538    | 0.363%  |
| IGMP (2)    | 11,994    | 0.100%  |
| ICMPv6 (58) | 5,602     | 0.047%  |

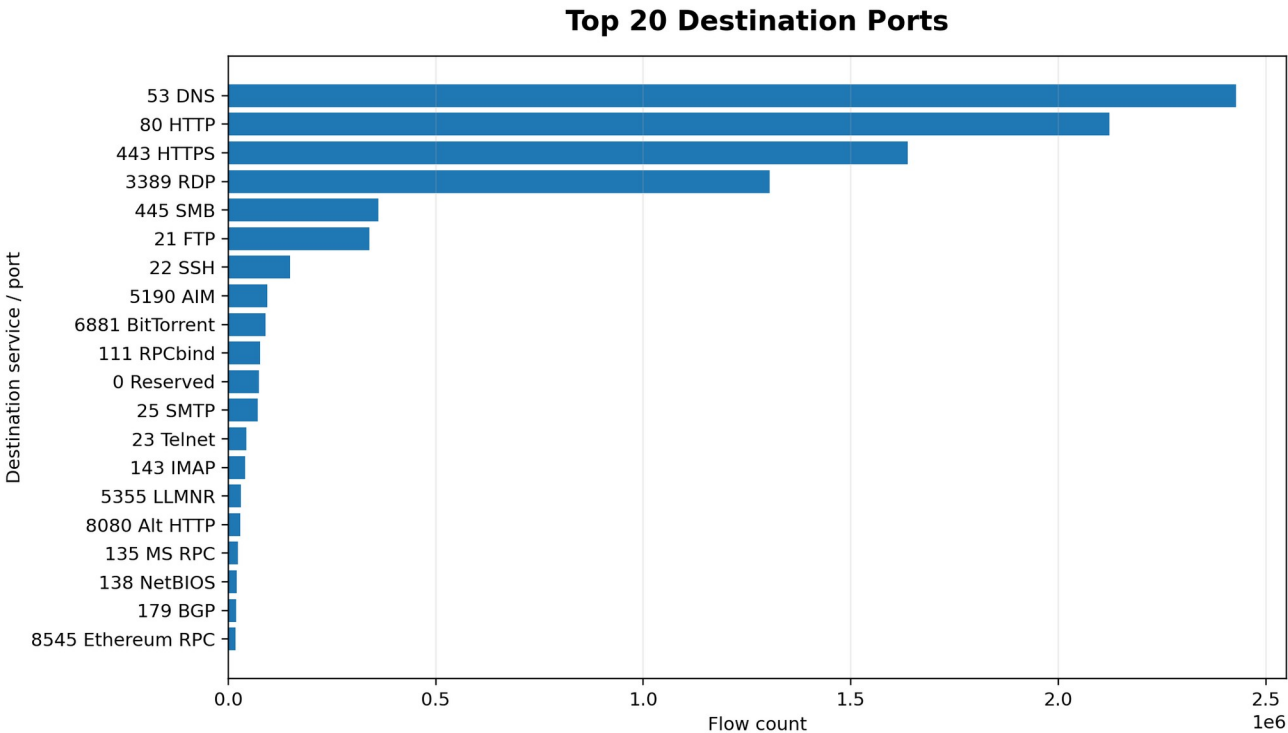


Figure 5. Top 20 destination ports. DNS, HTTP, HTTPS, and RDP are the largest service groups.

### Security interpretation

The port profile includes common user services and frequently attacked services such as RDP (3389), SMB (445), FTP (21), SSH (22), Telnet (23), and Windows RPC/NetBIOS ports. Port 0 can represent non-TCP/UDP traffic or exporter-specific missing-port encoding.

## 6. Numerical Feature Statistics

| Feature     | Mean      | Minimum | Median | Maximum     |
|-------------|-----------|---------|--------|-------------|
| L4_SRC_PORT | 41,245.19 | 0       | 50,610 | 65,535      |
| L4_DST_PORT | 8,964.76  | 0       | 443    | 65,535      |
| PROTOCOL    | 8.66      | 0       | 6      | 255         |
| L7_PROTO    | 19.32     | 0       | 0      | 251         |
| IN_BYTES    | 3,983.73  | 0       | 232    | 228,223,480 |
| OUT_BYTES   | 9,489.45  | 0       | 156    | 243,219,688 |
| IN_PKTS     | 35.63     | 0       | 4      | 1,221,043   |
| OUT_PKTS    | 12.52     | 0       | 2      | 1,349,068   |
| TCP_FLAGS   | 50.96     | 0       | 24     | 223         |

|                                |           |   |           |           |
|--------------------------------|-----------|---|-----------|-----------|
| FLOW_DURATION_MILLI<br>SECONDS | 2,163,429 | 0 | 4,178,245 | 4,294,967 |
|--------------------------------|-----------|---|-----------|-----------|

### Distribution pattern

Byte and packet features have medians far below their means and extremely large maxima, indicating strong right skew. The flow-duration maximum is close to  $2^{32} / 1000$  milliseconds, suggesting an exporter or timeout boundary near 71.6 minutes.

## 7. Distribution Visualizations

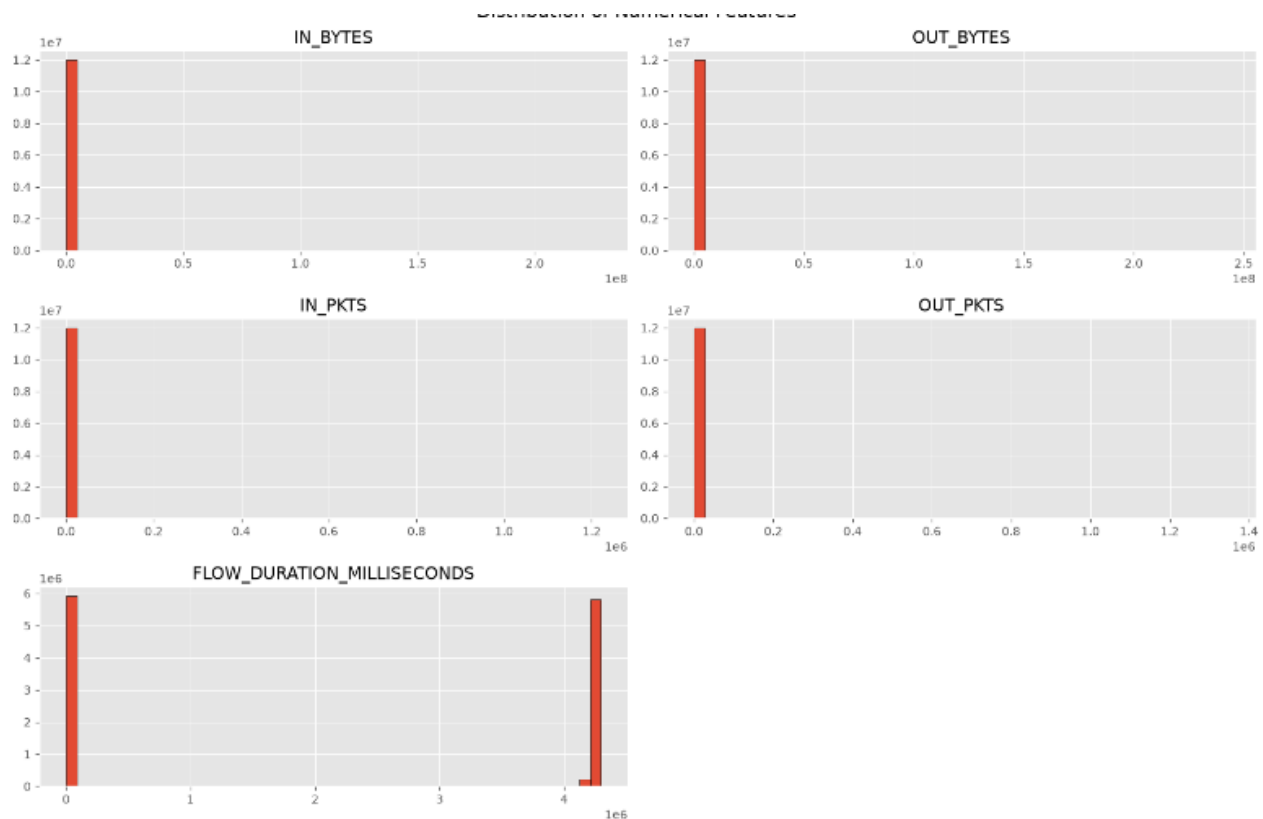
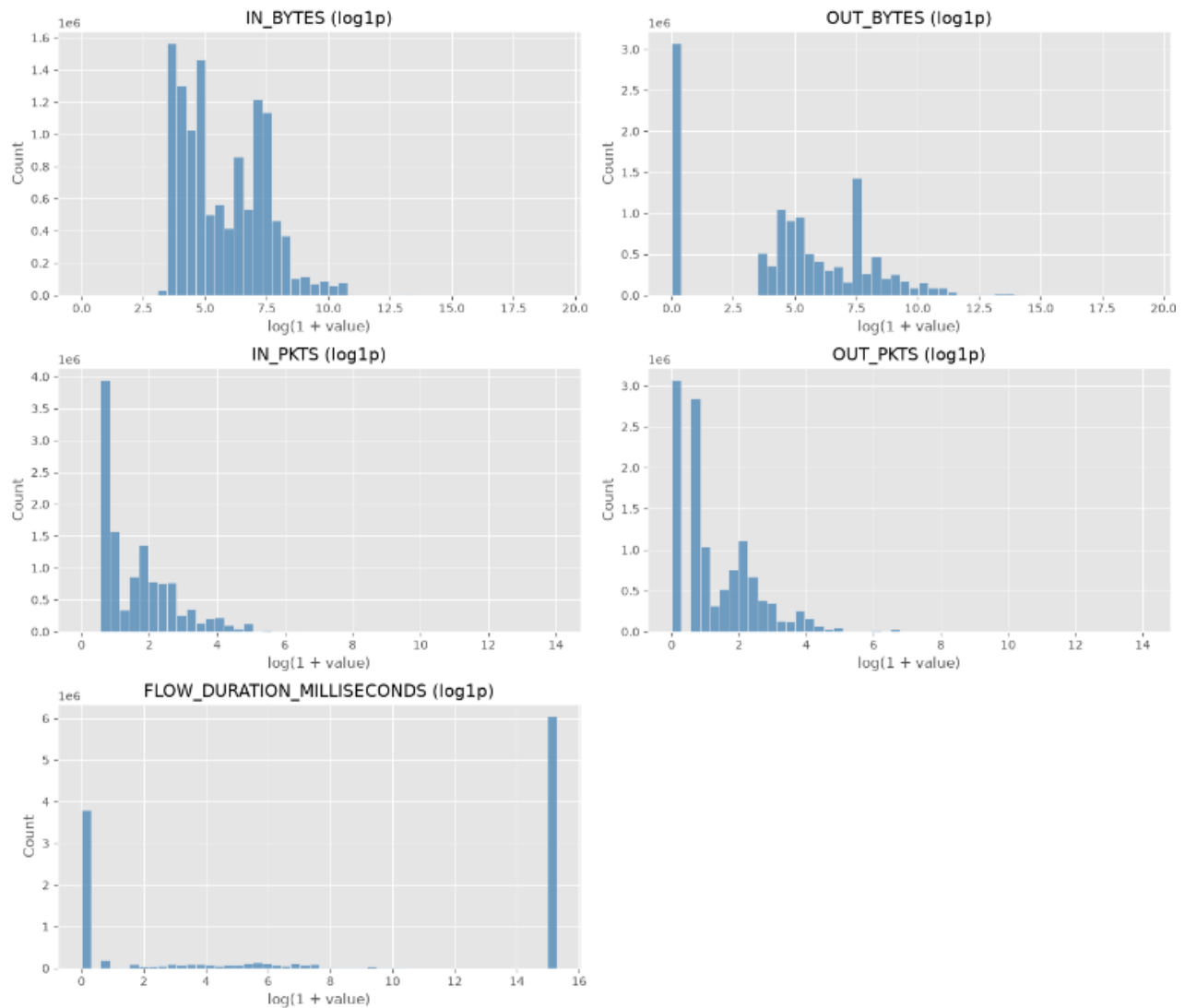


Figure 6. Raw-scale histograms drawn in the notebook. Extreme values compress most observations into the first bins, while flow duration displays strong concentrations near zero and near the upper boundary.

### Why the raw plots look compressed

A small number of very large byte and packet values determine the x-axis range. This makes the dense region of ordinary flows difficult to inspect and motivates a logarithmic transformation for visualization.



*Figure 7. Log1p-transformed histograms drawn in the notebook. The transformation  $\log(1+x)$  compresses the long tail while safely retaining zero-valued observations.*

### Interpretation of log1p

The transformation changes the viewing scale, not the underlying observations. It reveals multiple traffic clusters, large zero spikes in outgoing traffic, and persistent bimodality in flow duration. Whether log1p should also be used during model preprocessing must be validated separately.

## 8. Relationship and Class Analysis

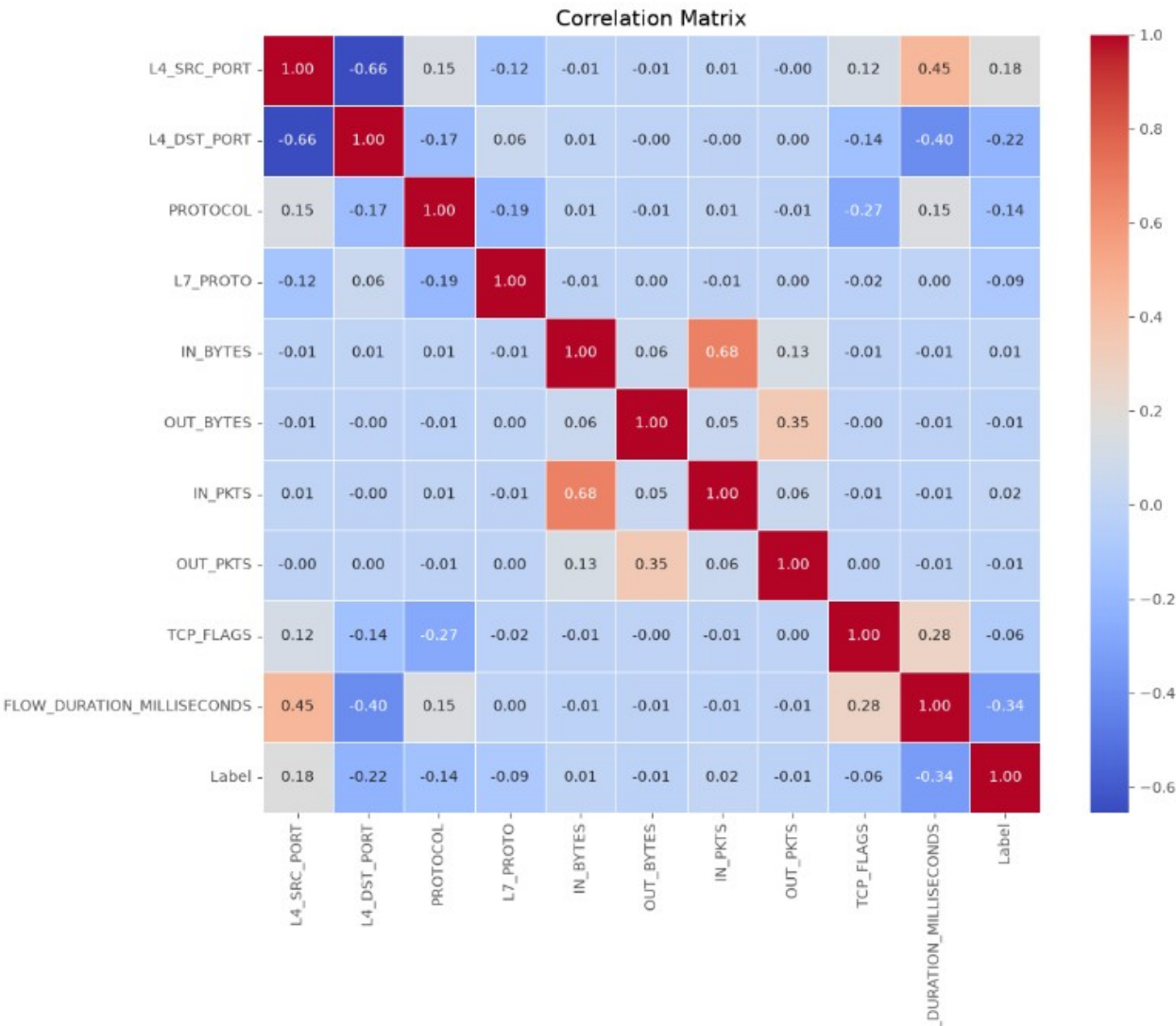


Figure 8. Pearson correlation matrix drawn in the notebook.

| Relationship               | Correlation | Interpretation                                                                     |
|----------------------------|-------------|------------------------------------------------------------------------------------|
| IN_BYTES vs IN_PKTS        | +0.68       | More incoming packets usually produce more incoming bytes, but packet size varies. |
| L4_SRC_PORT vs L4_DST_PORT | -0.66       | High ephemeral client ports commonly connect to lower well-known service ports.    |
| OUT_BYTES vs OUT_PKTS      | +0.35       | Moderate relationship due to variation in outgoing packet sizes.                   |
| FLOW_DURATION vs Label     | -0.34       | Attack and benign flows have meaningfully different duration patterns.             |
| L4_DST_PORT vs Label       | -0.22       | Destination service selection contributes some linear signal.                      |

## Correlation limitation

Pearson correlation measures linear association only. Weak correlation does not imply that a feature is useless; the model may learn nonlinear interactions among ports, protocols, flags, duration, packets, and bytes.

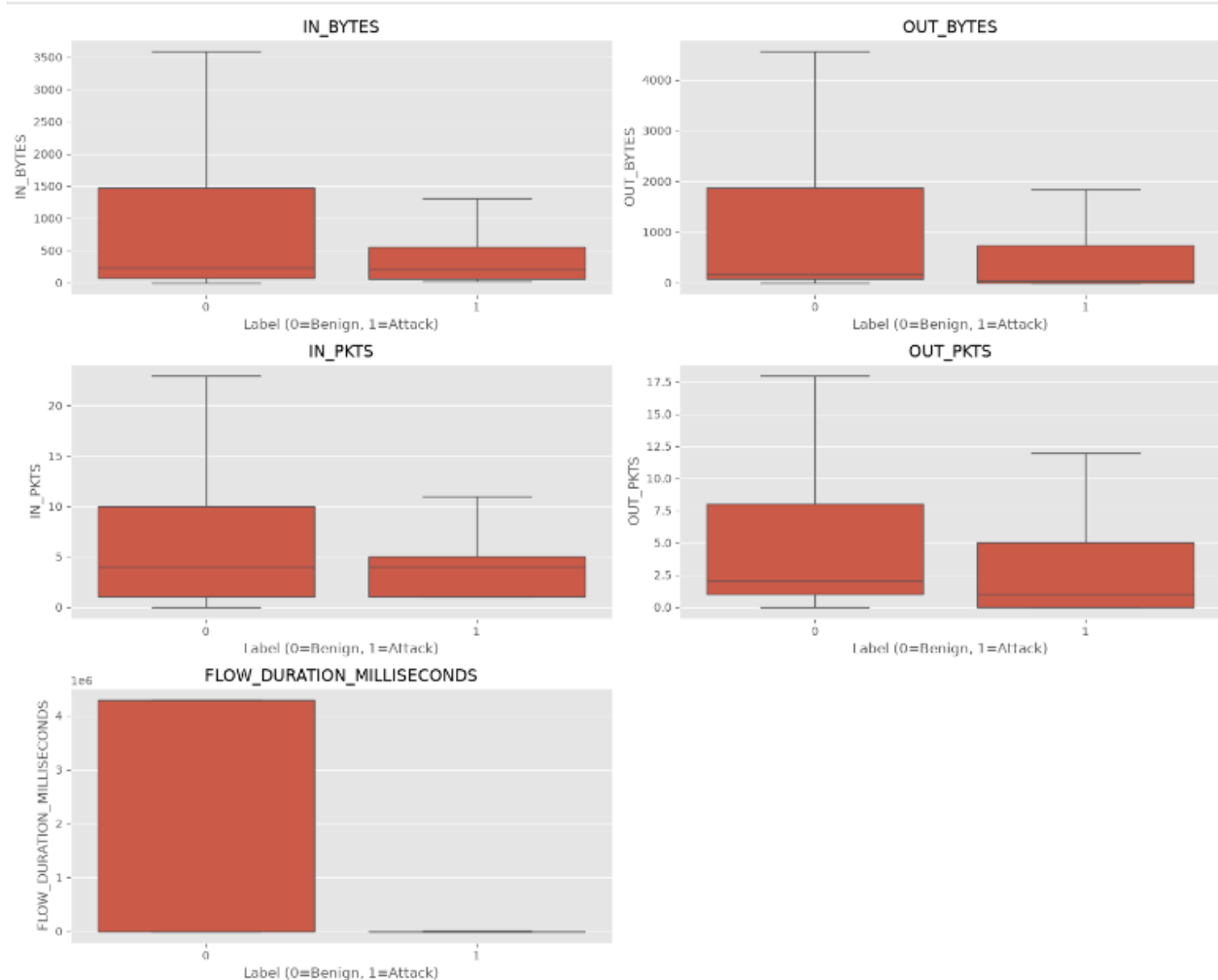


Figure 9. Boxplots comparing numerical features by binary label, with extreme fliers hidden for readability.

## Class-separation finding

Benign flows generally show wider byte and packet distributions, while attack flows are more concentrated. The classes still overlap strongly, confirming that no single threshold is sufficient and that multivariate modeling is necessary. Flow duration remains difficult to read on the raw scale and is better studied after transformation.

## 9. TCP Flag Distribution

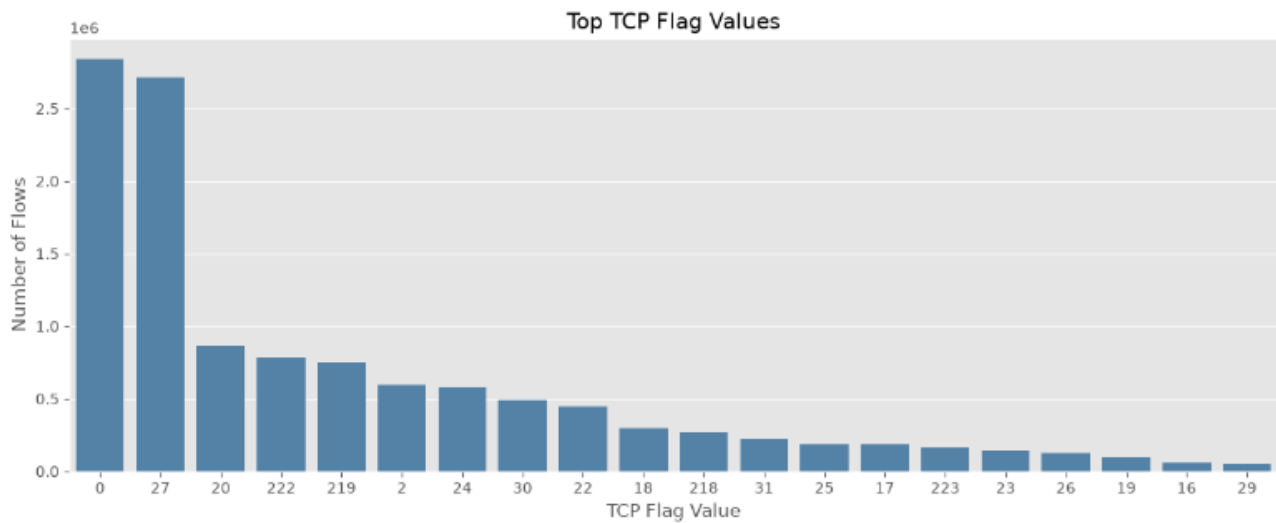


Figure 10. Most frequent aggregated TCP flag values drawn in the notebook.

### Bitmask interpretation

TCP\_FLAGS is an aggregated bitmask rather than an ordinal measurement. Values such as 27 can encode combinations of FIN, SYN, PSH, and ACK observed across a flow. The feature should be retained, but its integers should not be interpreted as a simple continuous magnitude.

## 10. IP Address Cardinality and Leakage Risk

| IP feature    | Unique values | Most frequent example | Frequency |
|---------------|---------------|-----------------------|-----------|
| IPV4_SRC_ADDR | 75,630        | 172.31.69.25          | 840,864   |
| IPV4_DST_ADDR | 27,826        | 172.31.0.2            | 2,045,759 |

| Top source IP   | Count   | Top destination IP | Count     |
|-----------------|---------|--------------------|-----------|
| 172.31.69.25    | 840,864 | 172.31.0.2         | 2,045,759 |
| 192.168.1.31    | 269,925 | 172.31.69.25       | 940,292   |
| 192.168.1.30    | 268,910 | 192.168.1.190      | 287,600   |
| 192.168.1.36    | 244,364 | 192.168.100.3      | 272,805   |
| 192.168.1.33    | 195,761 | 192.168.1.184      | 240,988   |
| 18.221.219.4    | 193,360 | 169.254.169.254    | 238,785   |
| 192.168.100.148 | 171,096 | 18.221.219.4       | 193,360   |

|            |         |               |         |
|------------|---------|---------------|---------|
|            |         |               |         |
| 59.166.0.5 | 153,762 | 192.168.1.195 | 191,669 |
| 59.166.0.1 | 153,439 | 192.168.1.152 | 190,226 |
| 59.166.0.4 | 153,404 | 149.171.126.1 | 153,450 |

**Decision: remove both IP columns**

IP addresses identify hosts in the benchmark environments and can enable memorization of attack sources or targets. Removing them encourages the model to learn transferable traffic behavior and reduces leakage risk in zero-day evaluation.

11. Final EDA Conclusions

| Finding               | Conclusion                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Data completeness     | All 15 columns are complete; no missing-value imputation is required.                                                 |
| Duplicates            | 10.55% exact duplicates require controlled analysis before removal and must not be split across training and testing. |
| Binary imbalance      | The 3.3:1 benign-to-attack ratio is moderate, but accuracy alone is misleading.                                       |
| Multiclass imbalance  | Attack classes follow a severe long-tail distribution; rare-class and zero-day evaluation must be designed carefully. |
| Source imbalance      | NF-CSE-CIC-IDS2018 dominates the merged corpus, so per-source evaluation is valuable.                                 |
| Feature distributions | Bytes and packets are highly right-skewed, with legitimate extreme values and many zero observations.                 |
| Feature redundancy    | No numerical pair has $ r  > 0.90$ ; all behavioral features can be retained initially.                               |
| Generalization        | Source and destination IP addresses should be removed to reduce memorization and environment dependence.              |

12. Recommended Preprocessing Plan

| Step | Action                                 | Reason                                            |
|------|----------------------------------------|---------------------------------------------------|
| 1    | Remove IPV4_SRC_ADDR and IPV4_DST_ADDR | Prevent memorization and improve transferability. |

|   |                                                    |                                                                                            |
|---|----------------------------------------------------|--------------------------------------------------------------------------------------------|
| 2 | Preserve Attack and Dataset metadata               | Required for zero-day holdouts and source-stratified evaluation.                           |
| 3 | Resolve duplicates before splitting                | Avoid duplicate leakage across train and test.                                             |
| 4 | Create zero-day and ordinary train/test partitions | All held-out attack categories must be absent from training.                               |
| 5 | Fit transformations on training data only          | Prevent preprocessing leakage.                                                             |
| 6 | Scale numerical features                           | Neural networks benefit from comparable feature scales.                                    |
| 7 | Evaluate log1p plus scaling as an experiment       | Strong skew may benefit from transformation, but it should be validated.                   |
| 8 | Use imbalance-aware evaluation/training            | Report per-class metrics and consider class weighting or resampling only on training data. |

### Final assessment

NF-UQ-NIDS is a large, diverse, and suitable benchmark for a Deep RL based IDS. Its main methodological risks are duplicate leakage, environment-specific IP memorization, source dominance, and severe attack-category imbalance. The completed EDA provides direct evidence for each preprocessing decision.